BEYOND ACCURACY

A DEEP INVESTIGATION INTO FAILURE MECHANISMS OF CLASSIFICATION MODELS

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INTRODUCTION

Why Accuracy Is Not Enough?

- Accuracy is the most used metric — but often misleading
- A model can score 99% and still be fundamentally broken
- High accuracy can hide:
  - Data leakage
  - Bias
  - Memorization
- Real-world failure can be catastrophic — especially in healthcare, law, and finance.

THE BASELINE ILLUSION

“When a Dumb Model Wins”

- A Dummy Classifier predicts only the most frequent class
- In imbalanced datasets, it can achieve surprisingly high accuracy
- Example: 90% of patients are healthy →

predicting "healthy" always = 90% accuracy

Key takeaway:

Always compare your model against a simple baseline before trusting its results

DATA LEAKAGE

“The Silent Cheat”

Two types:

- Target Leakage — training data contains hints of the answer
- Train-Test Contamination — test data leaks into training

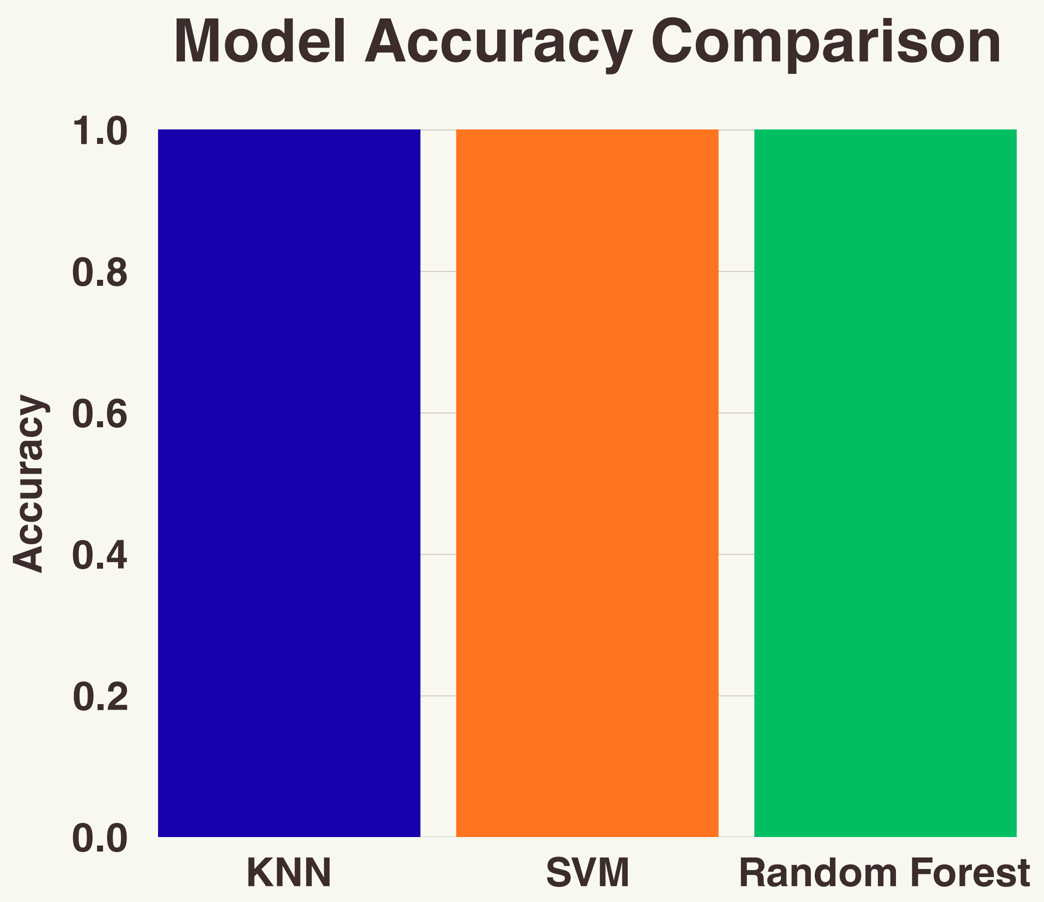
Why it's dangerous:

- Creates false confidence in model performance
- The flaw only appears after deployment
- By then — the cost is high

DATA LEAKAGE — CASE STUDY

Disease Prediction: A Real Example

- Dataset: 4,920 records → 4,616 were duplicates (94%)
- Duplicate patterns increased the risk of data leakage
- Result: All models achieved 100% accuracy



How We Addressed:

- Removed duplicated records
- Used stratified train-test split
- Evaluated with precision, recall, F1-score, and confusion matrix

Recommendation:

External validation is needed before trusting the model in real-world scenarios.

LABEL NOISE

- Label noise = incorrect labels in training data
- Causes: human error, poor collection, system failures

Effects on models:

- Learns wrong patterns
- Lower prediction accuracy
- Leads to overfitting

MEMORIZATION VS. LEARNING

Does the Model Truly Understand?

Memorization:

* Stores specific training examples
* Perfect on training data — fails on new data
* Common when dataset is too small or model too complex

True Learning:

* Recognizes meaningful patterns
* Generalizes to unseen data
* Performs reliably in real-world scenarios

→ High training accuracy ≠ good model

ADVERSARIAL EXAMPLES

One Pixel Changes Everything

* Small, invisible changes to input can completely fool a model
* The model classifies incorrectly with high confidence

Real-world risks:

* Self-driving cars misreading stop signs
* Facial recognition systems bypassed
* Medical imaging misclassified

Why it happens:

* Models rely on irrelevant features
* Lack of data diversity
* Gap between lab and real-world conditions

ALGORITHMIC BIAS

”The Model Is Biased — And Doesn't Know It“

How bias enters data:

* Representation bias — underrepresented groups
* Classification bias — human prejudice encoded into labels

Key insight:

High overall accuracy can mask unequal error rates across groups

Famous cases:

* COMPAS (2016) — racial bias in criminal risk scores
* Amazon Hiring Algorithm (2018) — discriminated against women
* Healthcare Algorithm (2019) — denied care to Black patients

CONCLUSION

Beyond the Number

Six hidden failure modes:

✗ Baseline Illusion — are you beating random guessing?

✗ Data Leakage — did the model cheat?

✗ Label Noise — is the data trustworthy?

✗ Memorization — did it learn or memorize?

✗ Adversarial Attacks — is it robust?

✗ Algorithmic Bias — is it fair?

Accuracy is a starting point — not a finish line

RECOMMENDDATIONS

Before Trusting Any Model — Ask:

Data:

* Was train-test split done before preprocessing?
* Are there duplicate records?
* Are labels accurate and unbiased?

Evaluation:

* Was a baseline compared?
* Were F1, precision, recall examined?
* Were subgroup error rates analyzed?

Deployment:

* Was the model tested on independent data?
* Is there a monitoring system post-deployment?

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THANK YOU

Questions and Discussion Welcome